

USE-CASE FLOW SPECIFICATION

UC-01 — Log Monthly Consumption

Primary Actor

Resident

Goal

Record monthly household consumption so the system can calculate and track monthly CO2e emissions.

Preconditions

- Resident is registered /authenticated.
- The selected month is available for entry.
- Required conversion factors are available.

Postconditions

- Monthly electricity, gas, and commute data are stored.
- Monthly CO2e is calculated.
- Historical emissions and reduction-target progress are updated.

Main Success Scenario

1. Resident selects Log Monthly Consumption.
2. System displays fields for electricity, gas, and commute kilometers.
3. Resident enters the monthly values.
4. System validates the entered values.
5. System stores the monthly consumption record.
6. System calculates monthly CO2e using the specified conversion factors.
7. System updates the historical emissions record.
8. System updates reduction-target progress.
9. System displays the saved record, CO2e result, and updated progress.

Alternate Flow — Invalid Input

- 3a. A value is missing, non-numeric, negative, or otherwise invalid.
- 3a1. System rejects it and identifies the affected field.
- 3a2. Resident corrects the value and submits again.
- 3a3. System resumes validation and continues the main flow.

Alternate Flow — Conversion Factor Unavailable

- 6a. Required conversion factors are unavailable.
- 6a1. System informs the Resident and does not produce a misleading zero/negative CO2e value.
- 6a2. Resident can retry when factors are available.